MARPAC RDIMS # 1111764

ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Submarine Escape and Rescue Coordinator Course

NQUAL - AILF

MITE Course ID - 122827

Chapter 1&2 – 04 Apr 23
Chapter 3&4 – 10 Oct 19
Version 5.0

RECORD OF AMENDMENTS

The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre (Atlantic) (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments are to be requested through NTDC(A) and approved by CNPTG.

Revision	Approval Date	Authority
Original	14 Jan 98 18 Dec 97	MS TRG Det 4500-1-Sub (Chap 1&2) FMFCS:4500-1(SMER) (CHAP 3&4)
One	Conversion to VIC Class QSP Board (OCT 02)	DMTE (Chap 1&2) CFNOS (Chap 3&4)
QSP-MS: 1.00	QSP Board (18 Nov 05) Chap1&2(01 Mar 06)	DMTE (Chap 1&2)
QSP-MS: 2.00	23 Nov 06	DMTE (Chap 1&2)
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QSP-MS: 3.01	11 Aug 08	DMTE 2-8
QSP-MS: 4.00	29 Sep 10	DMTE (Chap 1&2)
QSP-MS: 4.01	1 Oct 10	DMTE (Chap 3&4)
5.0		C NPTG

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FOREWORD

1. This training is based on the specification found at:

   a. A-PD-055-002/PP-001;
   b. A-PD-055-002/PP-002; and
   c. A-PD-055-003/PQ-OSS.

A trainee will achieve the qualification AILF when the standards listed in all Performance Objectives have been met.

2. The designated TE for this training are indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A)	All	122827	AILF

CHAPTER 1

TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training while performing the role of a Submarine Escape and Rescue Coordinator of a Victoria-class submarine – AILF. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22, para 4.5. For this course, a period of instruction is 50 minutes in duration with 8 periods per day.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	96
Enabling Checks (ECs)	21
Performance Checks (PCs)	34
Actual Training Time (ATT) Sub-total	151.0 **18.8 Trg Days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	2.0
Training Administrative Time (TAT)	2.0
Training Travel Time (TTT)	0
Physical Fitness (PT)	18
Sub-total	2.75 3 **Trg Days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) / 8 * **Partial days must be rounded up**	21.625 22 **Trg Days**

REQUIRED COURSE STAFFING

Instructional Staff

POSITION	RANK	QUALIFICATIONS	QTY	PO / EO
Course Officer	Lt(N)	MS Eng or NCS Eng AILS, ALDE or HOD	1	001 - 002
Senior Instructor	CPO2	AILF	1	001 - 002

Guest Lecturers

POSITION	RANK	QTY	PO / EO
National SUBSAR Coord	CPO2	1	EO 001.01
Psychologist - Traumatic Stress Expert	CAF or Civ	1	EO 002.07
Sea Training (Submarines) Coxswain	CPO1	1	PC 002
Submarine / Dive Medical Officer	Lt(N) - LCdr	1	EO
RJOC Watch Officer	Lt(N) - LCdr	1	EO 002.07
CSF OPS	Lt(N) - LCdr	1	EO 001.01- 03
Coastal SUBSAR Coordinator	PO1 - CPO2	1	EO 001.01
FDU (A) / (P)	PO1 - CPO2	1	EO 001.01

CRITICAL TRAINING RESOURCES

FACILITY	NO. OF DAYS
Alongside Submarine	13
SUBSAR Reaction Stores	0.5
SUBSAR Emergency Life Support Stores	0.5
Regional Joint Operations Centre Joint Rescue Coordination Centre	0.5
Training Aids for simulation purposes to include: a. Training PLB b. Training Pyros c. Training Comm Buoys d. UWT e. Escape suits f. Red Cross Locker contents IAW A1 g. LiOH training Curtains h. Atmospheric monitoring equipment i. EO2-26H O2 candle with generator j. MROD k. First Aid Kit l. Pressure tight Pod and Receiving Bag m. Long BU's. n. EBS Mask's o. Mannequin p. Victoria Class Guard Cards	22

COURSE CAPACITY

3. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 8;

   b. Minimum – 4; and

   c. Optimum – 6.

PREREQUISITES

To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. AILS;

b. RQ-PO1 or QL6B;

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

Nil.

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 Coordinate Submarine Escape and Rescue Training for DISSUB Survivors

1. Performance. Coordinate Submarine Escape and Rescue Training for DISSUB Survivors

2. Conditions.

a) Given:

(1) Submarine with functional escape compartment or trainer;

(2) Submarine Crew Publications and Drawings;

(3) Victoria Class Guard Cards;

(4) References;

(5) Training Scenarios; and

(6) Submarine Escape and Rescue related equipment and systems.

b) Denied: Assistance.

c) Environmental: Both at Sea and Alongside Submarine.

3. Standard. The Submarine Escape and Rescue Coordinator shall, IAW references, coordinate Submarine Escape and Rescue training for submarine crews by:

a) Advising DISSUB Survivors on the Roles of Submarine Escape and Rescue Organizations at Local, National and International Levels to include:

(1) Local – Formation and Regional Level:

(a) SUBSAR Reaction Stores;
(b) SUBSAR Emergency Life Support Stores;
(c) Regional Joint Operations Centre;
(d) Joint Rescue Coordination Centre;

(2) National Level:

(a) Canadian Joint Operations Command;
(b) CSF Readiness – National SUBSAR Coordinator;
(c) CSF Operations – SUBOPAUTH;
(d) Submarine Escape and Rescue Advisory Team;
(e) DISSUB Liaison Team;

(3) International Level:

(a) ISMERLO;

b) Advising DISSUB Survivors on Submarine Accountability (Operational Messages) to include:

(1) Submarine Notice (SUBNOTE);

(2) Submarine Check Report (SUBCHECK);

(3) Submarine Safety Communication Check (COMCHECK);

(4) Surfacing Zero Time (SZER);

(5) Diving Signal; and

(6) Submarine Surfacing Signal.

c) Advising DISSUB Survivors on Indications of a Possible Submarine Accident to include:

(1) SUBLOOK;

(2) SUBMISS; and

(3) SUBSUNK.

d) Advising DISSUB Survivors on Submarine Escape and Rescue Methods and Procedures to include:

(1) Rescue; and/ or

(2) Escape.

e) Surveying Submarine Escape Compartment Equipment during a DISSUB in accordance with the Submarine Escape and Rescue Inspection to include:

(1) Mustering all Escape and Rescue related equipment; and

(2) Rigging all Escape and Rescue related equipment.

f) Operate the Single Escape Tower during a DISSUB in accordance with the Tower Functional Trial to include:

(1) Conducting unmanned runs of the Single Escape Tower.

g) Describing Medical Considerations Associated with DISSUB Survivors to include:

(1) Physiological and environmental factors to include:

(a) DISSUB internal pressure;
(b) DISSUB atmosphere;
(c) Temperature;
(d) Hydration;
(e) Nutrition;
(f) Management of Survivors;
(g) Management of causalities; and
(h) Management of cadavers.

4. References. All references utilized IAW Annex A.

5. Task Numbers. T0001, T0002, T0003, T0004, T0005, T0006, T0007, T0008, T0009

6. Training Limitations. Availability of a submarine or trainer with a functional escape and rescue suite is critical in order to coordinate DISSUB Survivors – Submarine Escape and Rescue training to meet the standard for this PO.

EO 001.01(A) – Advise DISSUB Survivors on the Roles of Submarine Escape and Rescue Organizations at Local, National and International Level

Conditions		Sequence
Classroom		EO 001.01
Time	**Classification**	**Instructor**
8 X 50 Minute Periods	UNCLAS	AILF National SUBSAR Coord

Learning Objectives	Method
Roles of National Submarine Escape and Rescue Organizations: 1. Canadian Joint Operations Command 2. SERWG 3. SERC; 4. CSF Readiness – National SUBSAR Coordinator; 5. CSF Operations – SUBOPAUTH; 6. Submarine Escape and Rescue Advisory Team; 7. DISSUB Liaison Team; Roles of International Submarine Escape and Rescue Organizations: 1. ISMERLO; 2. NATO SMERWG 3. Asia Pacific Submarine Conference; and 4. STANAG's.	Interactive Lecture Brainstorm

Scenario	National and International Supporting Agencies response in the event of a DISSUB.

Activity Steps	Introduction to PO 001 (Brainstorm) As this PO is directly related to training the crew in a DISSUB situation – the instructor will start off asking the students what important points they think every Submariner should know – writing them on a whiteboard. (10 min) The instructor will introduce the Guest Lecturer. The Guest Lecturer will go over the LOs covered in EO 001.01 and explain the focus will be on the students being able to present the material to a submarine crew in a DISSUB situation. The Guest Lecturer will reinforce the importance of understanding this material as it relates directly to reassuring DISSUB Survivors of what the response of supporting agencies is going to be.

Resources	**References**
1. EO 001.01(A) PPT provided by Guest Lecturer 2. Guest Lecturer is advised 3. CJOC CONPLAN 4. ISMERLO Web Site	1. CFCD 103 2. ATP 10 3. ATP 57 4. CJOC CONPLAN 5. ISMERLO Web Site

Notes to Instructor
Students should be very familiar with each organization by end of lectures.

EO 001.01(B) – Advise DISSUB Survivors on the Roles of Submarine Escape and Rescue Organizations at Local, National and International Level

Conditions		Sequence
In-Situ		After EO 001.01(A)
Time	**Classification**	**Instructor**
12 X 50 Minute Periods	UNCLAS	Guest Lecturer (Unit POC's)

Learning Objectives	Method
1. Roles of Local Submarine Escape and Rescue Organizations within MARLANT / MARPAC to include: a. SUBSAR Reaction Capability and Emergency Life Support Stores (4 periods); b. RJOC and JRCC (4 periods); and c. FDU (A) / (P) (4 periods).	Conduct the following tours: 1. SUBSAR Reaction and Emergency Life Support Stores. 2. RJOC 3. JRCC 4. FDU

Scenario	Local Agencies response in the event of a DISSUB.
Activity Steps	Instructor will arrange for tours of the local agencies listed above. This includes making logistical arrangements, security, and for a unit POC to conduct the tour emphasizing their role in a DISSUB situation. Additionally, request that the Unit POC link the National and International agencies to their respective organizations. The instructor will emphasis the importance of understanding this material as it relates directly to reassuring DISSUB Survivors of what the response of supporting agencies is going to be.

Resources	References
1. 1 X 9 passenger van 2. Security arrangements	1. CFCD 103 2. ATP 10 3. ATP 57 4. CJOC CONPLAN 5. ISMERLO Web Site

Notes to Instructor
Students should be very familiar with each organization by end of lectures.

EO 001.02 – Advising DISSUB Survivors on Submarine Accountability (Operational Messages)

Conditions		Sequence
Classroom		After EO 001.01(B)
Time	**Classification**	**Instructor**
2 x 50 minute period	UNCLAS	CANSUBFOR Ops

Learning Objectives	Method
1) Describe Submarine Accountability – Operational Messages and Effective Communications to include: a. Submarine Notice (SUBNOTE); b. Submarine Check Report (SUBCHECK); c. Submarine Safety Communication Check (COMCHECK); d. Surfacing Zero Time (SZER); e. Diving Signal; and f. Submarine Surfacing Signal.	Interactive Lecture

Scenario	The failure to receive any of the above expected transmitted signal from a submarine may well be the very first indication that a submarine is experiencing difficulties.
Activity Steps	1. Instructor will introduce each subject and present lecture via PPT and provide the class with sample messages. 2. After each topic is presented, instructor will provide a scenario of when that particular message is transmitted. 3. After each message has been explained, the instructor will ask students questions as confirmation that the material has been well understood.

Resources	References
1. EO 001.02 PPT provided by CSF Ops 2. Sample Messages	1. CFCD 103 2. CJOC SUBSAR CONPLAN

Notes to Instructor

It is essential that each message is clearly defined and a confirmation question shall be asked to various students.

EO 001.03 – Advising DISSUB Survivors on Indications of a Possible Submarine Accident

Conditions		Sequence
Classroom		After EO 001.02
Time	**Classification**	**Instructor**
2 x 50 minute period	UNCLAS	CANSUBFOR Ops

Learning Objectives	Method
1) Describe the various procedures and phases that occur after the following code words are initiated by CSF: a. SUBLOOK; b. SUBMISS; and c. SUBSUNK. 2) Explain the indication of a Submarine accident.	Interactive Lecture

Scenario	Code words of the procedures initiated by CSF when the safety of a submarine is in doubt.
Activity Steps	1. Instructor will introduce subject and present lecture via PPT. 2. After each topic is presented, instructor will present a scenario of when each message will be used. 3. After each message has been explained, the instructor will ask students questions as confirmation that the material has been understood.

Resources	References
1. EO 001.03 PPT provided by CSF Ops 2. Sample Messages	1. CFCD 103 2. CJOC SUBSAR CONPLAN

Notes to Instructor
It is essential that each indication is clearly defined and a confirmation question shall be asked to various students.

EC 001.03 – Advise DISSUB Survivors

Scope	Conditions	Sequence
EO 001.01-03	Classroom	After EO 001.03
Time	**Classification**	**Instructor**
1 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.01-03. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. EC 001.03 will be a written test based on content from EO 001.01, 001.02 and 001.03. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test	1. CFCD 103 2. ATP 10 3. ATP 57 4. CJOC CONPLAN 5. ISMERLO Web Site

Notes to Instructor
N/A

EO 001.04(A) – Advise DISSUB Survivors on Submarine Escape and Rescue Methods and Procedures

Conditions		Sequence
Classroom		After EO 001.03
Time	**Classification**	**Instructor**
8 x 50 minute period	UNCLAS	AILF

Learning Objectives	Method
1. Describe rescue methods utilizing both the Submarine Rescue Vehicle and Submarine Rescue Chamber to include: A. Capabilities and limitations; B. Types and origin of rescue vehicles; C. Response times; D. Intervention Systems; E. Previous mating and de-mating exercises (i.e. Dynamic) 2. Describe the use of the Victoria Class Guard Cards with regards to accepting Rescue and Intervention Systems (Cards 19 – 21).	Interactive Lecture Table Top exercise

Scenario	Survivors of a DISSUB evacuating the submarine by either a Rescue Vehicle or Rescue Chamber
Activity Steps	1. Instructor will introduce subject and present lecture via PPT and ISMERLO Web Site. 2. After each topic is presented, instructor will present a scenario (Victoria Class Guard Card Table Top Exercise) of when one of the type of Rescues Systems and Intervention Systems would be utilized. 3. Students to be broken into groups in order to respond to a DISSUB scenario provided by the instructor. This will include a viable Rescue system and the rationale behind their decision making process.

Resources	References
1. EO 001.04 PPT 2. ISMERLO Web Site 3. Victoria Class Guard Cards X 8	1. Victoria Class Guard Cards 2. CFCD 103 3. ISMERLO Web Site

Notes to Instructor
It is essential that each topic is clearly defined and a confirmation question shall be asked to various students.

EO 001.04(B) – Advice DISSUB Survivors on Submarine Escape and Rescue Methods and Procedures

Conditions		Sequence
Classroom		After EO 001.03(A)
Time	**Classification**	**Instructor**
4 x 50 minute period	UNCLAS	AILF

Learning Objectives	Method
Describe Single Tower Escape considerations to include: a) Capabilities and limitations; b) Atmosphere (Contaminated or Uncontaminated) c) Survival on the surface d) Medical considerations e) Describe the use of the Victoria Class Guard Cards with regards to utilizing Single Tower Escape (Cards 1 – 3) (Cards 6 – 7).	Interactive Lecture Table top

Scenario	Escaping from a DISSUB utilizing the Single Escape Tower
Activity Steps	1. Instructor will introduce subject and present lecture via PPT. 2. After the topic is presented, instructor will present a scenario of when the Single Tower Escape method could be used IAW Victoria Class Guard Cards. 3. Students to be broken into groups in order to respond to a DISSUB scenario provided by the instructor. This will include scenarios which force the students to make the decision to effect a Single Tower Escape, rather than await a Rescue System.

Resources	References
1. EO 001.04 PPT 2. Victoria Class Guard Cards X 8	1. Victoria Class Guard Cards 2. CFCD 103 3. ISMERLO

Notes to Instructor
It is essential that each topic is clearly defined and a confirmation question shall be asked to various students.

EC 001.04 – Advise DISSUB Survivors on Submarine Escape and Rescue Methods and Procedures

Scope	Conditions	Sequence
EO 001.04	Classroom	After EO 001.04(B)
Time	**Classification**	**Instructor**
1 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.04. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. EC 001.04 will be a written test based on content from EO 001.04. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test 4. Victoria Class Guard Cards	1. CFCD 103 2. ATP 10 3. ATP 57 4. CJOC CONPLAN 5. ISMERLO Web Site **6.** Victoria Class Guard Cards

Notes to Instructor
N/A

EO 001.05 – Survey Submarine Escape Compartment Equipment during a DISSUB in accordance with the Submarine Escape and Rescue Inspection

Conditions		Sequence
Submarine/Trainer		After EO 001.04
Time	**Classification**	**Instructor**
8 x 50 minute period	UNCLAS	AILF

Learning Objectives	Method
IAW the Victoria Class Guard Cards and Escape and Rescue Inspection, locate all Escape and Rescue related equipment.	Practical demonstration

Scenario	Mustering Escape and Rescue Equipment during a DISSUB
Activity Steps	1. Upon giving an overview, the instructor will have the students identify the stowage location of all items IAW the Escape and Rescue Inspection.

Resources	References
1. Working copy of the Escape and Rescue Inspection X 8 2. Victoria Class Guard Cards X 8 3. Submarine or Trainer	1. Victoria Class Guard Cards 2. CFCD 103 3. Escape and Rescue Inspection ISMERLO

Notes to Instructor
It is essential that all escape and rescue equipment is clearly defined and a locations confirmed on the Submarine/Trainer. Question shall be asked to various students for confirmation.

EC 001.05 – Survey Submarine Escape Compartment Equipment during a DISSUB in accordance with the Submarine Escape and Rescue Inspection

Scope	Conditions	Sequence
EO 001.05	Classroom	After EO 001.05
Time	**Classification**	**Instructor**
1 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.05. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. EC 001.05 will be a written test based on content from EO 001.05. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test 4. Victoria Class Guard Cards 5. Escape and Rescue Inspection	1. CFCD 103 2. Victoria Class Guard Cards **3.** Escape and Rescue Inspection

Notes to Instructor
N/A

EO 001.06 – Operate the Single Escape Tower during a DISSUB in accordance with the Tower Functional Trial

Conditions		Sequence
Submarine/Trainer		After EO 001.05
Time	**Classification**	**Instructor**
8 x 50 minute period	UNCLAS	AILF

Learning Objectives	Method
Operate the Single Escape Tower IAW Tower Function Trial	Review or test and trial reports Practical demonstration

Scenario	DISSUB
Activity Steps	1. Instructor will show the Single Escape Tower to the students and then discuss trial and agenda reports, and provide practical demonstration utilizing the Tower Functional Trial. 2. The students will be broken into groups in order to perform simulated runs IAW the Tower Function Trial and Victoria Class Guard Cards (Cards 4 and 5).

Resources	References
1. Tower Function Trials working copies 2. Single Escape Tower/Trainer 3. Victoria Class Guard Cards	1. Victoria Class Guard Cards 2. Tower Function Trials 3. CFCD 103

Notes to Instructor

It is essential that each topic is clearly defined and a confirmation question shall be asked to various students.

* Student are NOT expected to complete a full Tower Functional Trial (i.e.: air, water and at sea runs). However, students' are expected to have an in-depth knowledge of the Single Escape Tower operating procedures.

EC 001.06 – Operate the Single Escape Tower during a DISSUB in accordance with the Tower Functional Trial

Scope	Conditions	Sequence
EO 001.06	Classroom	After EO 001.06
Time	**Classification**	**Instructor**
1 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.06. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. EC 001.06 will be a written test based on content from EO 001.06. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test 4. Victoria Class Guard Cards 5. Tower Function Trial	1. CFCD 103 2. Victoria Class Guard Cards **3.** Tower Function Trial

Notes to Instructor
N/A

EO 001.07 – Describe Medical Considerations Associated with DISSUB Survivors

Conditions		Sequence
Classroom		After EO 001.06
Time	**Classification**	**Instructor**
4 x 50 minute period	UNCLAS	Guest Lecturer: Sub Dive Med Officer

Learning Objectives	Method
IAW guest lecturer's brief, describe medical, physiological and environmental factors associated with DISSUB Survivors to include: a) DISSUB internal pressure; b) DISSUB atmosphere; c) Temperature; d) Hydration; e) Nutrition; f) Management of Survivors; g) Management of causalities; and h) Management of cadavers.	Interactive Lecture Group Discussions

Scenario	Casualty management during DISSUB
Activity Steps	IAW Guest Lecturer's curriculum

Resources	References
1. Book guest	1. IAW Guest Lecturer's curriculum

Notes to Instructor
It is essential that each topic is clearly defined and a confirmation question shall be asked to various students.

EC 001.07 – Describe Medical Considerations Associated with DISSUB Survivors

Scope	Conditions	Sequence
EO 001.07	Classroom	After EO 001.07
Time	**Classification**	**Instructor**
1 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.07. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. EC 001.07 will be a written test based on content from EO 001.07. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test 4. Victoria Class Guard Cards	1. CFCD 103 2. Victoria Class Guard Cards

Notes to Instructor
N/A

PC 001 – Coordinate Submarine Escape and Rescue Training for DISSUB Survivors

Scope	Conditions	Sequence
PO 001	Classroom	After EC 001.07
Time	**Classification**	**Instructor**
2 X 50 Mins	UNCLAS	AILF

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.07. Exam shall contain short answer questions and may contain multiple choice. <u>INSTRUCTIONS GIVEN TO TRAINEE</u> 1. PC 001 will be a written test based on content from PO 001. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1. Classroom 2. One test per student 3. DS Solution for test 4. Victoria Class Guard Cards 5. Escape and Rescue Inspection 6. Tower Function Trial	1. CFCD 103 2. ATP 10 3. ATP 57 4. CJOC CONPLAN 5. ISMERLO Web Site 6. Victoria Class Guard Cards 7. Escape and Rescue Inspection 8. Tower Function Trial

Notes to Instructor
N/A

PO 002 Execute Duties as a Senior Survivor in a DISSUB

1. Performance. Execute Duties of a Submarine Senior Survivor in a DISSUB

2. Conditions.

a) Given:

(1) Submarine or trainer;

(2) Submarine Crew Publications and Drawings;

(3) Victoria Class Guard Cards;

(4) References;

(5) Training Scenarios; and

(6) Submarine Escape and Rescue related equipment and systems.

3. Standard. The Submariner shall, IAW Victoria Class Guard Cards, execute duties of Senior Survivor during a DISSUB by:

a) Preparing the Escape Compartment for Escape or Rescue by:

(1) Carrying out Initial Actions (Card 1 and Card 3) to include:
- Determining if the atmosphere is breathable and courses of action;
- Conducting rapid assessment of injured personnel;
- Surveying damage to the submarine and courses of action;

(2) Commencing Immediate Tower Escape (Card 2) to include:
- Delegation of the following:
  - Preparing the Tower (Card 4);
  - Shutting down the bulkhead;
  - Rigging two BU's near the Tower;
- Compartment pressure considerations and courses of action (Card 3);
- Preparing the Tower (Card 4) to include:
  - Tower preps
  - Unmanned runs courses of action
  - Conduct unmanned run (Card 5)

(3) Implementing DISSUB procedures for awaiting Rescue (Card 3) to include:
- Nomination of 2nd Senior Survivor;
- Adhering to the Senior Survivor's Guide (Card 15) to include:
  - Developing a survivors plan;
  - Inspection of equipment;
  - Personnel considerations;
  - Lighting;
  - Position Indications;
  - Deceased personnel;
  - Situation Reports;
- Assigning the following teams:
  - Tower Operating (Card 4 and Card 5);
  - Water / Rations / Stores (Card 16);
  - Communication (Cards 17 to 20);
- Assigning Atmosphere Monitoring Team to include:
  - Atmosphere monitoring instructions (Card A);
  - Partial Pressure Correction (Card B);
  - Atmosphere Control for Carbon Dioxide (Card C);
  - Atmosphere Control for Oxygen (Card D);
  - Chlorine Monitoring Instructions (Card E);
  - Atmosphere Record Sheets (Card F and Card G);
  - Air Purification Stores Location (Card H)

b) Preparing the Single Escape Towers to accept (Intervention) Emergency Life Support Stores – POD Delivery, IAW Victoria Class Guard Cards by:

(1) Rigging equipment (Card 21; and
(2) Employing Communication Codes (Card 21);

c) Evacuate the Submarine by Rescue in accordance with Victoria Class Guard Cards to include:

(a) Draining down the SRV seat cofferdam and hatch cavity
(b) Equalising the pressure between the DISSUB and RESCUE VEHICLE after it is sealed in position on the seat
(c) Shutting the hatch cavity drain valves once the upper hatch is open (ATP hatch)
(d) Shut the tower flood hull valve and tower flood valve once the upper hatch is open (AFT)
(e) Mating rescue vehicle
(f) Directing rescue vehicle boarding;
(g) Evacuating casualties, and
(h) De-mating rescue vehicle.

4. References. A3, A8, A34

5. Task Numbers. T0005, T0009

6. Training Limitations. Availability of a submarine or trainer with a functional escape and rescue suite is critical in order to coordinate DISSUB Survivors – Submarine Escape and Rescue training to meet the standard for this PO.

EO 002.01 – Prepare the Escape Compartment for Submarine Escape and Rescue in accordance with Victoria Class Guard Cards

Conditions		Sequence
Submarine or Trainer		Start EO 002.01
Time	**Classification**	**Instructor**
16 X 50	UNCLAS	AILF

Learning Objectives	Method
1. Carry out Initial Actions (Card 1 and Card 3). 2. Commence Immediate Tower Escape (Card 2). 3. Implement DISSUB procedures for awaiting Rescue (Card 3).	Practical Interactive Role Playing Explain – Demo - Imitate

Scenario	DISSUB – Escape and/or Rescue
Activity Steps	1. Prior to this evolution – the instructor shall conduct a safety brief to the students, supporting staff, and role players. 2. Instructor shall take students onboard the submarine or trainer and demonstrate how they should set up the escape compartment for DISSUB conditions IAW Victoria Class Guard Cards. 3. Once all the students are satisfied with the demo – the instructor will appoint one of the students as the Senior Survivor rotating through the entire class. This will ensure that all students are provided an opportunity to be the Senior Survivor. 4. Each of the remaining students or role players (depending on the course load) will act as members of the various supporting teams and casualties. 5. This will include scenarios which force the students to make the decisions.

Resources	References
1. Submarine; or 2. Trainer 3. Escape and Rescue Suite; 4. Victoria Class Guard Cards X 8 5. Role Players X 7 (depending on the course load) 6. Training Scenarios	1. VCSOP 2. CFCD 103 3. Victoria Class Guard Cards

Notes to Instructor
Each Student shall be familiar with how to employ Victoria Class Guard Cards.

EO 002.02 – Prepare the Single Escape Towers to accept (Intervention) Emergency Life Support Stores – POD Delivery

Conditions		Sequence
Submarine or Trainer		EO 002.02 after 002.01
Time	**Classification**	**Instructor**
8 X 50 mins	UNCLAS	AILF

Learning Objectives	Method
1. Prepare the Single Escape Towers to accept (Intervention) Emergency Life Support Stores – POD Delivery, IAW Victoria Class Guard Cards; and 2. Receive Emergency Life Support Stores	Practical Interactive Role Playing

Scenario	DISSUB – Intervention Systems
Activity Steps	1. Prior to this evolution – the instructor shall conduct a safety and communication brief to the students, Supporting staff, and role players. 2. Instructor shall take students onboard the submarine or trainer and demonstrate how they should set up the single escape tower to accept (Intervention) Emergency Life Support Stores and receive a POD Delivery during a DISSUB. 3. The students shall set up the single escape tower to accept (Intervention) Emergency Life Support Stores and receive a POD Delivery during a DISSUB IAW Victoria Class Guard Cards and the given scenarios. 4. Supporting staff (using safety comms – simulating an UWT) will be on the Submarine Casing providing Hatch Control while the POD is inserted into the Tower.

Resources	References
1. Submarine; or 2. Trainer 3. Escape and Rescue Suite; 4. Victoria Class Guard Cards X 8 5. Role Players X 7 (depending on the course load) 6. Training Scenarios 7. Communication devices 8. Exercise POD 9. Tools for tapping codes	1. CFCD 103 2. Victoria Class Guard Cards

Notes to Instructor
Each Student shall be familiar with how to employ Victoria Class Guard Cards.

EO 002.03 – Evacuate the Submarine by Rescue in accordance with Victoria Class Guard Cards

Conditions		Sequence
Submarine or Trainer		EO 002.03 after 002.02
Time	**Classification**	**Instructor**
16 X 50 mins	UNCLAS	AILF

Learning Objectives	Method
(1) Carry out Rescue Vehicle procedures IAW Victoria Class Guard Cards (Card 19 and Card 20).	Practical Interactive Role Playing

Scenario	DISSUB - Evacuation
Activity Steps	1. Prior to this evolution – the instructor shall conduct a safety brief to the students and supporting staff. 2. Instructor shall take students onboard the submarine or trainer and demonstrate how they should set up the escape compartment for rescue from a DISSUB IAW Victoria Class Guard Cards. 3. Once all the students are satisfied with the demo – the instructor will appoint one of the students as the Senior Survivor rotating through the entire class. This will ensure that all students are provided an opportunity to be the Senior Survivor. 4. Each of the remaining students or role players (depending on the course load) will act as casualties. 5. This will include scenarios which force the students to make the decisions.

Resources	References
1. Submarine; or 2. Trainer; 3. Escape and Rescue Suite; 4. Victoria Class Guard Cards X 8; 5. Role Players X 7 (depending on the course load); 6. Training Scenarios; 7. Communication devices; and 8. Tools for tapping codes.	1. CFCD 103 2. Victoria Class Guard Cards

Notes to Instructor
Each Student shall be familiar with how to employ Victoria Class Guard Cards.

EC 002.03 – Evacuate the Submarine by Escape or Rescue in accordance with Victoria Class Guard Cards

Scope	Conditions	Sequence
EO 002.01-03	Submarine / Trainer	After EO 002.03
Time	**Classification**	**Instructor**
2 X 50 Min (per student)	UNCLAS	AILF

Scenario	DISSUB scenario provided by the instructor with multiple injects.
Assessment Steps	1. Prior to this EC – the instructor shall conduct a safety brief to the students and supporting staff. 2. The student is provided all resources and the escape compartment. 3. The student shall react to queues provided by the instructor IAW Victoria Class Guard Cards. 4. The instructor will provide each student with a scenario and inject events/ situations in order to make the students react accordingly. 5. To evacuate as many submariners as possible by either escaping or being rescued from a DISSUB. 6. Debrief students on the results of their assessment.

Resources	References
1. Submarine; or 2. Trainer; 3. Escape and Rescue Suite; 4. Victoria Class Guard Cards; 5. Role Players X 7 (depending on the course load); 6. Training Scenarios; 7. Communication devices 8. Training POD; and 9. Tools for tapping codes.	1. CFCD 103 2. Victoria Class Guard Cards

Notes to Instructor
Arrange for role players and training resources. During this EC students are to act as role players.

Limitations	N/A
Impact of Failure	N/A

Item	EC 002.03 – Evacuate the Submarine by Escape or Rescue in accordance with Victoria Class Guard Cards	SAT	UNSAT	Remarks
1. Preparing the Escape Compartment for Escape or Rescue by:				
Carry out Initial Actions (Card 1) to include:				
Determining if the atmosphere is breathable and courses of action		☐	☐	
Surveying damage to the submarine and courses of action		☐	☐	
Determining Compartment pressure		☐	☐	
Determine next course of action		☐	☐	
2. Commencing Immediate Tower Escape (Card 2) to include:				
Preparing the Tower (Card 4)		☐	☐	
Shutting down the bulkhead		☐	☐	
Rigging two BU's near the Tower		☐	☐	
Compartment pressure considerations and courses of action (Card 3)		☐	☐	
Preparing the Tower (Card 4) to include: - Tower preps - Unmanned runs courses of action - Conduct unmanned run (Card 5)		☐	☐	
3. Implementing DISSUB procedures for awaiting Rescue (Card 3) to include:				
Nomination of 2nd Senior Survivor		☐	☐	
Adhering to the Senior Survivor's Guide (Card 15) to include: - Developing a survivors plan; - Inspection of equipment; - Personnel considerations; - Lighting; - Position Indications; - Deceased personnel; - Situation Reports;		☐	☐	
Assigning the following teams: - Tower Operating (Card 4 and Card 5); - Water / Rations / Stores (Card 16); - Communication (Cards 17 to 20);		☐	☐	
Assigning Atmosphere Monitoring Team to include: - Atmosphere monitoring instructions (Card A); - Partial Pressure Correction (Card B); - Atmosphere Control for Carbon Dioxide (Card C); - Atmosphere Control for Oxygen (Card D); - Chlorine Monitoring Instructions (Card E);		☐	☐	

- Atmosphere Record Sheets (Card F and Card G); - Air Purification Stores Location (Card H)			
4. Preparing the Single Escape Towers to accept (Intervention) Emergency Life Support Stores – POD Delivery, IAW Victoria Class Guard Cards by:			
- Rigging equipment (Card 21)	☐	☐	
- Employing Communication Codes (Card 21)	☐	☐	
5. Evacuate the Submarine by Rescue IAW Victoria Class Guard Cards to include:			
- Draining down the SRV seat cofferdam and hatch cavity	☐	☐	
- Shutting the hatch cavity drain valves once the upper hatch is open (ATP hatch)	☐	☐	
- Shut the tower flood hull valve and tower flood valve once the upper hatch is open (AFT)	☐	☐	
- Mating rescue vehicle	☐	☐	
- Directing rescue vehicle boarding	☐	☐	
- Evacuating casualties, and	☐	☐	
- De-mating rescue vehicle	☐	☐	

Score: ___ / 22

For this EC, the student shall achieve SAT on all factors relevant to their scenario.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002 – Execute Duties as a Senior Survivor in a DISSUB

Scope	Conditions	Sequence
PO 002	Submarine / Trainer	After EC 002.03
Time	**Classification**	**Instructor**
4 X 50 Min (per student)	UNCLAS	AILF

Scenario	DISSUB scenario provided by the instructor with multiple injects.
Assessment Steps	1. Prior to this PC – the instructor shall conduct a safety brief to the students and supporting staff. 2. The student is provided all resources and the escape compartment. 3. The student shall react to queues provided by the instructor IAW Victoria Class Guard Cards. 4. The instructor will provide each student with a scenario and inject events/ situations in order to make the students react accordingly. 5. To evacuate as many submariners as possible by either escaping or being rescued from a DISSUB. 6. Debrief students on the results of their assessment.

Resources	References
1. Submarine; or 2. Trainer; 3. Escape and Rescue Suite; 4. Victoria Class Guard Cards; 5. Role Players X 7 (depending on the course load); 6. Training Scenarios; 7. Communication devices 8. Training POD; and 9. Tools for tapping codes.	1. CFCD 103 2. Victoria Class Guard Cards

Notes to Instructor
Arrange for role players and training resources. During this PC students are **NOT** to act as role players.

Limitations	N/A
Impact of Failure	IAW NPTORD 4500-54

Item	PC 002 – Execute Duties as a Senior Survivor in a DISSUB	SAT	UNSAT	Remarks
1. Preparing the Escape Compartment for Escape or Rescue by:				
Carry out Initial Actions (Card 1) to include:				
Determining if the atmosphere is breathable and courses of action		☐	☐	
Surveying damage to the submarine and courses of action		☐	☐	
Determining Compartment pressure		☐	☐	
Determine next course of action		☐	☐	
2. Commencing Immediate Tower Escape (Card 2) to include:				
Preparing the Tower (Card 4)		☐	☐	
Shutting down the bulkhead		☐	☐	
Rigging two BU's near the Tower		☐	☐	
Compartment pressure considerations and courses of action (Card 3)		☐	☐	
Preparing the Tower (Card 4) to include: - Tower preps - Unmanned runs courses of action - Conduct unmanned run (Card 5)		☐	☐	
3. Implementing DISSUB procedures for awaiting Rescue (Card 3) to include:				
Nomination of 2nd Senior Survivor		☐	☐	
Adhering to the Senior Survivor's Guide (Card 15) to include: - Developing a survivors plan; - Inspection of equipment; - Personnel considerations; - Lighting; - Position Indications; - Deceased personnel; - Situation Reports;		☐	☐	
Assigning the following teams: - Tower Operating (Card 4 and Card 5); - Water / Rations / Stores (Card 16); - Communication (Cards 17 to 20);		☐	☐	
Assigning Atmosphere Monitoring Team to include: - Atmosphere monitoring instructions (Card A); - Partial Pressure Correction (Card B); - Atmosphere Control for Carbon Dioxide (Card C); - Atmosphere Control for Oxygen (Card D); - Chlorine Monitoring Instructions (Card E); - Atmosphere Record Sheets (Card F and Card G);		☐	☐	

- Air Purification Stores Location (Card H)			
4. Preparing the Single Escape Towers to accept (Intervention) Emergency Life Support Stores – POD Delivery, IAW Victoria Class Guard Cards by:			
- Rigging equipment (Card 21)	☐	☐	
- Employing Communication Codes (Card 21)	☐	☐	
5. Evacuate the Submarine by Rescue IAW Victoria Class Guard Cards to include:			
- Draining down the SRV seat cofferdam and hatch cavity	☐	☐	
- Shutting the hatch cavity drain valves once the upper hatch is open (ATP hatch)	☐	☐	
- Shut the tower flood hull valve and tower flood valve once the upper hatch is open (AFT)	☐	☐	
- Mating rescue vehicle	☐	☐	
- Directing rescue vehicle boarding	☐	☐	
- Evacuating casualties, and	☐	☐	
- De-mating rescue vehicle	☐	☐	

Score: ___ / 22

Trainee shall achieve SAT all factors in order to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Ref Code	Publication
A1	C-22-594-000/NA-001 Escape Equipment Inspection (VCS)
A2	B-GN-002-000/RQ-001 CFCD 102 Maritime Command Combat Readiness Requirements
A3	C-23-VIC-000/MS-001 VICTORIA Class Submarine Guard Cards
A4	CJOC – CONPLAN EXPEDITIONARY SUBMARINE SEARCH AND RESCUE (E-SUBSAR)
A5	VICTORIA Class Submarine Standing Orders (VCSSOs)
A6	Submarine Licensing and Certification Management System (SLCMS) dated 9 Dec 02
A7	VICTORIA Class Emergency Operating Procedures (VCEOPs)
7A8	C-23-SUB-002/MS-001 CFCD 103 Submarine Escape and Rescue Manual
A9	C-27-040-000/NT-A01 Tower Functional Trial
A10	VICTORIA CLASS SUBMARINE READINESS TRAINING GUIDE (VCSRTG)
A11	C-87-003-000/SG-001 Purity of Compressed Breathing Air and Gases for Divers
A12	C-87-020-001/NG-001 Special Test Instructions - Diving Breathing Air Analysis
A13	D-87-003-000/SG-001: CF Standard – Purity of Compressed Breathing Air and Gases for Divers
A14	C-51-456-000/MS-001 Type 639 Submarine Indicator Buoy / Type 639 Indicator Unit
A15	C-28-020-001/TB-001 Technical Bulletin (Marine) for Testing of Shipboard Lifting Appliances
A16	PMS E104/28 Emergency Fresh Water
A17	PMS E105/04 Battery ATP Shipping Trunk (ATP/DSRV Trunk)
A18	PMS E105/05 DSRV and Cofferdam
A19	PMS E107/01 Pressure Hull Hatches
A20	EO2-26H Self Contained Oxygen Generator Technical Article
A21	PMS E515/01 Lifting Appliances and Fittings
A22	PMS E517/05 Absolute Pressure Gauge
A23	PMS E614/01 BIBS / HIS
A24	PMS E614/02 Indicator Buoy Release Mechanisms
A25	PMS E614/03 and L614/03 Escape Tower Fittings
A26	PMS E614/04 Escape Compartments
A27	PMS L502/14 Dragger
A28	PMS W105/10 SSE Mk 9
A29	PMS W201/01 Pyro Lockers
A30	PMS W201/59 Sonar Type 2073 (Emergency UWT)
A31	PMS C-51-456-000/NY-001 Type 639 Indicating Unit
A32	C-29-537-000/MS001 Breathing Air Quality Manual for Victoria Class
A33	Submarine Emergency Position Indicating Radio Beacon (SEPIRB) Maintenance, Operation and Instruction Manual
A34	VICTORIA Class Standard Operating Procedures (VCSOPs)
B1	ATP 10 NATO Search and Rescue Manual
B2	ATP/MTP 57 NATO Submarine Rescue Manual
B3	BR 2217(9) (4B) SSE Mk 9
B4	BR 8827 Underwater Telephone (UWT) Type 2073
B5	Lithium Hydroïde Curtains Technical Article
C1	International Submarine Escape and Rescue Liaison Organization website: www.ISMERLO.org

A CF or other official Canadian government publication
B Official Allied Force or Allied government publication
C Commercial publication
D Other

	Task Statement	T/NT	PO
T0001	Conduct test and trials with outside agencies	T	
T0002	Conduct escape equipment inspection	T	
T0003	Conduct post inspection procedures	T	
T0004	Conduct pre inspection procedure	T	
T0005	Conduct escape /rescue training	T	
T0006	Review Planned Maintenance Schedules	T	
T0007	Review Test and Trial Reports	T	
T0008	Coordinate load-out of SUBSAR reaction equipment	T	
T0009	Operate Emergency Life Support Stores (ELSS)	T	

Annex C – QSP WORKING GROUP
RECORD OF PROCEEDINGS

Dates: 18-21 April 2017, 10-21 July 2017.

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Bryson	D	CPO1	00114	CSF
Jacobs	P	CPO1	00366	ST(S)
McGuire	P	CPO1	00114	HMCS Windsor
MacPhee	B	CPO2	00299	CSF
Bergeron	R	CPO2	00324	NTDC(A)
Girard	M	CPO2	00324	NFS(A)
Lilly	B	CPO2	00366	ST(S)
Lockley	C	CPO2	00299	HMCS Chicoutimi
Johnston	D	PO1	00366	CSF
Arbour	R	Civ.	n/a	Marlant FTA
Perry	ME	Capt	00211	NTDC(A)

Working Group Objective:

1. To develop Submarine Senior Survivor Course to address the need of the Canadian Submarine Force. Chapter 1 and 2 were addressed first with follow up to take place for chapters 3 and 4.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation

Discussions / Decisions:

1. The Board convened under the direction of the Submarine Escape and Rescue Committee Chair on behalf of the CRCN to better reflect the duties and responsibilities of a Senior Survivor onboard the submarine.

2. Board decided to cancel the OJPR portion of the existing course, due to no added benefit in training and the administration burden it created. CPO2 Bergeron will action during post board activities.

3. Board de-linked the SUBSAR duties and responsibilities from the course. The duties and responsibilities of Reaction Stores, Life Support Stores and SITS will be moved to the Combat Systems Engineering Chief Course

4. Board changed the name of the course to Senior Survivor Course vice Submarine Escape and Rescue Co-ordinator course, to better reflect the duties and responsibilities of the Senior Survivor.

5. Board finished chapter 1 and 2 and re-convened 10 – 21 July in order to complete chapters 3 and 4.

6. Abandonment of Ship was discussed. The board decided that because it falls under VCEOP #3 and is a whole boat evolution it should remain there, as it is not a primary responsibility of the Senior Survivor but rather a Command/HQ1 function.

7. Board took time to research the requirement of sea survival training for the Senior Survivor and sub community as a whole. Delta in Sub Training has been discovered. Board recommends that an amendment be made to VCEOP 3 and Victoria Class Guard Cards 6 and 7 to have sea survival actions on the surface to be added to the tower escape brief.

8. CSF will investigate the requirement for surface sea survival training to align with the fleet.

9. Board discussed conducting the Pilot Course on the West Coast due to the availability of the HMCS Victoria to be used as the training platform. This would also be a great opportunity to test the Modularization concept for future bi-coastal training. It was identified that the instructor will have to come from NFS(A), Escape and Rescue Inspector (West or East Coast) and TDO support for pilot evaluation can come from either TDC(A) or TDC(P).

10. CSF is exploring a study to upgrade the Weapon Handling and Discharge System Trainer (WHDST) to include an escape tower fitted for rescue and escape training. The vision of the board is that this will alleviate the need for an operational submarine as a critical training resource to run the course.

11. Investigate the possibility of using Ship Boarding and Cold Water Survival Course (1 day) delivered through Survival System Training Located in Dartmouth, NS, and have it modified to suit the VIC Class.

Lessons Learned:

12. There was too much emphasis on SUBSAR and not enough on the Senior Survivor part of the training. So this was to be changed in the creation of this training plan.

Guidance to Development and/or Instructional Staff:

13.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	**Cancel MITE Code for OJPR- switch it to Mod 2 for Course.**	**CPO2 Bergeron**	**NLT Nov 17**
2	**Conversation with IMQ for SETT availability for functional trials.**	**CPO2 Girard**	**NLT Nov 17**

Item	Action	OPI	Expected Date of Completion
3	**Amendment be made to the Guard Cards for cards 6 and 7 to have sea survival actions on the surface to be added to the tower escape brief**	**CPO2 Johnston**	**NLT Jan 18**
4	**Amendment be made to the VCEOP 3 to include sea survival brief.**	**CPO2 Lilly**	**NLT Nov 17**
5	**Scenario Development Template**	**CPO2 Girard**	**NLT Nov 17**
6	**Scenarios Completed with DS solution**	**CPO2 Girard**	**NLT Jan 18**
7	**Courseware (training aids)**	**CPO2 Bergeron/ CPO2 Girard**	**NLT Nov 17**
8	**Investigate software requirements for Guard Card amendments and reprinting.**	**CPO2 Johnston**	**NLT Nov 17**

Other Considerations (if applicable):

1. **PO 002 –** For the Pilot Course, both inspectors are required to instruct in the course

Dates: 23 – 27 September 2019.

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	Unit
Bryson	D	Lt(N)	CANSUBFOR
Guthrie	N	SLt	NTDCA
McGuire	P	CPO1	ST(SUBS)
Lilly	B	CPO2	CANSUBFOR
Johnston	D	CPO2	ST(SUBS)
Harvey	N	CPO2	NTDCA
Snelgrove	A	CPO2	NFSA

Working Group Objective:

1. To review the Submarine Escape and Rescue Coordinator Course QSP in order to:

    a. Expedite the completion of the QSP for NTDCA endorsement and NPTG approval; and

    b. Meet the deadline to conduct the course prior early October 2019 to meet CANSUBFOR Force Generation and Succession Planning objectives.

OJPR Not required for this course.

Acronym or Abbreviation	Term or Definition
ADS	Atmospheric Diving Suit
AILF	Course Designator for the Submarine Escape Coordinator Course
ASR	Submarine Rescue Ship (USN)
ATA	Atmosphere Absolute
ATP	Allied Tactical Publication
BIBS	Built-in Breathing System
BSQ	Basic Submarine Qualification
BU	Breathing Unit
CAGE	Cerebral Arterial Gas Embolism
CCFL	Commander Canadian Fleet Atlantic
CCFP	Commander Canadian Fleet Pacific
CDSM	Consultant in Diving and Submarine Medicine
CFCD	Canadian Forces Command Document
CFMO	Canadian Forces Medical Order
CFMWC	Canadian Forces Maritime Warfare Centre
NFS(A)	Naval Fleet School (Atlantic)
NFS(P)	Naval Fleet School (Pacific)
CCSF	Commander Canadian Submarine Force
CJOC	Canadian Joint Operations Command
CONOPS	Concept of Operations
COSPAS	Refer to SARSAT (COSPAS represents the Russian translation)
CRCN	Commander Royal Canadian Navy
DCS	Decompression Sickness
DCI	Decompression illness
D Div S	Director of Diving Safety
DISSUB	Distressed Submarine
DMEPM(SM)	Director Maritime Equipment Project Management (Submarines)
DMHS	Director Maritime Health Services
DMPOR	Director Maritime Policy, Operations and Readiness
DRDC	Defence Research and Development Canada
DWP	Docking Work Period
E AND R (or E&R)	Escape and Rescue
EC	Engineering Change
ECB	Expendable Communication Buoy
EDU	Experimental Diving Unit
EDWP	Extended Docking Work Period
ELSS	Emergency Life Support Stores
FDU	Fleet Diving Unit
FMF	Fleet Maintenance Facility
FMS	Foreign Military Sales
FSW (or FSW)	Feet of sea water
FTA	Fleet Technical Authority
GMDSS	Global Marine Distress Signalling System
GPS	Global Positioning System
HIS	Hood Inflation System

IPC	Illustrated Parts Catalogue
IOC	Introduction of Operational Capability
ISO	International Standards Organization
I.V.	Intra-venous
JTF(A)	Joint Task Force Atlantic
JTF(P)	Joint Task Force Pacific
LIVEX	Live Exercise
LiOH	Lithium Hydroxide Curtains
MARLANT	Maritime Forces Atlantic
MARPAC	Maritime Forces Pacific
MODUK	Ministry of Defence (UK)
MOG	Maritime Operations Group
MOSHIP	Mother Ship
MROD	Manual Reverse Osmosis Desalinator
msw	Metres of Sea Water
NATO	North Atlantic Treaty Organization
NDHQ	National Defence Headquarters
NSRS	NATO Submarine Rescue System
NPTG	Naval Personnel Training Group
NSN	NATO Stock Number
NTDC(A)	Naval Training Development Centre (Atlantic)
NTDC(P)	Naval Training Development Centre (Pacific)
OPI	Office of Primary Interest
OPORD	Operation Order
OPCOM	Operational Command
ORA	Operational Risk Assessment
OSC	On Scene Commander
OST	Operational Status Transfer
PCL	Pneumatic Components Limited
PLB	Personal Locator Beacon
POT	Pulmonary Oxygen Toxicity
pp	Partial Pressure
ppm	Parts per Million
PRM	Pressurized Rescue Module
QA	Quality Assurance
QAR	Quality Assurance Regional
RCC	Recompression Chamber
RN	Royal Navy
ROV	Remotely Operated Vehicle
SARSAT	Search and Rescue Satellite
SCV	Stole Charging Valve
SEIE	Submarine Escape Immersion Equipment
SERC	Submarine Escape and Rescue Committee
SEPIRB	Submarine Emergency Position Indicating Radio Beacon
SERWG	Submarine Escape and Rescue Working Group
SESSPE	Submarine Escape Surface Survival Personnel Equipment
SET	Single Escape Tower
SITREP	Situation Report
SLAB	Submarine Locating Acoustic Beacon

SLOT	Submarine Locating One-way Transmitter
SMASHEX	Submarine Accident Exercise
SMER	Submarine Escape and Rescue
SMERAT	Submarine Escape and Rescue Advisory Team
SMERWG	Submarine Escape and Rescue Working Group
SRC	Submarine Rescue Chamber
SRDRS	Submarine Rescue & Diving Recompression System
SRV	Submarine Rescue Vehicle
SSDR	Submarine Safety Docket Register
SSAS	Submarine Surface Abandonment Suit
SSE	Submerged Signal Ejector
SSTG	Submarine Sea Training Group
SSRA	Submarine Search and Rescue Authority
STANAG	NATO Standardization Agreement
STP	Standard Temperature Pressure
STO	Submarine Technical Officer
SUBOPAUTH	Submarine Operating Authority
SUBSAFE	Command Submarine Safety Office
SUBSAR	Submarine Search and Rescue
SUBSMASH	Submarine Accident
SW	Salt Water (or Sea Water)
USN	United States Navy
UWGTO	Underwater Group Technical Officer
UWT	Underwater Telephone
VOO	Vessel of Opportunity
WG	Water Gauge
WPET	Wet Pressurized Escape Training
WSC	Weapons Stowage Compartment

Annex F – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

Annex G – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: Sub Escape and Rescue Coord		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	8	$1,000.0	N/A	$8,000	$8,000	$40,000
Accommodations	8	$55.0	30	$13,200	$13,200	$66,000
Incidentals	8	$17.30	30	$4,153	$4,153	$20,765
Rations	8	$24.95	30	$5,988	$5,988	$29,940
Subtotal				**$31,341**	**$31,341**	**$156,705**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	8	$30.0	1	$240	$240	$1,200
Instructional Materials (Printed / Handouts etc)	8	$50.0	1	$400	$400	$2,000
Van Rental (7 passenger)	2	$250	4	$500	$500	$2,500
Subtotal				**$1,140**	**$1,140**	**$5,700**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (incremental)	1	258	30	$7,740	$7,740	$38,700
Subtotal				**$7,740**	**$7,740**	**$38,700**
	Total Estimated Costs			**$40,221**	**40,221**	**$201,105**

Annex H – COURSE HISTORY

1. This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.03	EO 001.01-03	To assess student's knowledge of DISSUB organization, operational messages, and indications.	Written	Pass mark is 70%
EC 001.04	EO 001.04	To assess student's knowledge of submarine escape and rescue methods and procedures.	Written	Pass mark is 70%
EC 001.05	EO 001.05	To assess the students ability to survey a submarine escape compartment in accordance with the submarine Escape and Rescue Inspection.	Written	Pass mark is 70%
EC 001.06	EO 001.06	To assess student's knowledge of Single Escape Tower operation during a DISSUB.	Written	Pass mark is 70%
EC 001.07	EO 001.07	To assess student's knowledge of DISSUB medical considerations.	Written	Pass mark is 70%
PC 001	PO 001	To assess student's knowledge related to coordinating DISSUB training for submariners.	Written	Pass mark is 70%
EC 002.03	EO 002.01-03	To assess student's competence in evacuating the submarine by escape or rescue using Victoria Class Guard Cards.	Practical	SAT on all EC checklist factors
PC 002	PO 002	To assess the student's competence executing the duties of a Senior Survivor in a DISSUB.	Practical	SAT on all EC checklist factors

Information about development of Technology Enabled Learning (TEL) can be found in the embedded document:

Spirals.docx